

Kunli Liu, PhD

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EDUCATION

University of Pennsylvania <i>Master of Science in Computer and Information Technology</i>	Philadelphia, PA 09/2019 – 05/2022
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Michigan State University <i>Ph.D. in Chemistry</i>	East Lansing, MI 08/2017 – 08/2022
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EXPERIENCE

Senior Research Scientist, Meta Platforms , Sunnyvale, CA <i>IC5 · Greatly Exceeded Expectations (top 10% performer)</i>	2022 – Present
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Unified E2E Test Platform — “TestPilot”

- Team lead of 10+ engineers from ads infra, ads products and ads change management to deliver unified E2E functional testing framework for ads infra
- Launched TestPilot, allowing developers to test the full ad lifecycle—from creation to delivery, accounting, and reporting—without deploying to production, the absence of which has resulted in significant inefficiencies in product development, increased reliability risks, and revenue loss of \$300M per year

Less Personalized Ads (LPA)

- Reliability area lead for LPA for enabling testing infrastructure to support LPA, which unblocked LPA launch and prevented \$10M revenue loss
- Launched observability framework for LPA which covers 100+ delivery-critical metrics at code change, service deployment and production stage.

Third-Party Data (3PD)

- Launched asset-level privacy policy enforcement with end-to-end propagation across the ads serving stack, unblocking the 3PD privacy commitment

Software Engineer Intern, Oracle Corporation	05/2021 – 08/2021
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- Launched an end-to-end testing flow for automated testing of Oracle Server Hardware via an event-driven mechanism.

GRADUATE RESEARCH ASSISTANT

Michigan State University , East Lansing, MI	08/2017 – 08/2022
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Hyaluronan Derivatization

- Developed a novel combinatorial library for hyaluronic acid derivatization via Ugi reactions, resulting in 6.5x more effective hyaluronic acid derivatives
- Employed machine learning and computational molecular modeling for predicting and evaluating binding of hyaluronan molecules.

SERS Image-Guided Surgery

- Developed one-pot synthetic methodology for Surface-Enhanced Raman Scattering (SERS) nanoparticles (NPs) with 2500x stronger signal compared to existing commercial SERS NPs..
- Built a machine-learning pipeline for real-time multimodal imaging reconstruction, demonstrating viability for intra-operative image-guided surgery.

PUBLICATIONS

1. **Liu, K.**, McCue, W.M., Yang, C.W., Finzel, B.C. and Huang, X. (2023). Combinatorial synthesis of a hyaluronan based polysaccharide library for enhanced CD44 binding. *Carbohydrate Polymers*, **300**, 120255. (Impact Factor: **12.5**)
2. **Liu, K.**, Ullah, A.A., Juhong, A., Yang, C.W., Yao, C.Y., Li, X., Bumpers, H.L., Qiu, Z. and Huang, X. (2024). Robust Synthesis of Targeting Glyco-Nanoparticles for Surface Enhanced Resonance Raman Based Image-Guided Tumor Surgery. *Small Science*, **4**(5), 2300154. (Impact Factor: **13.4**)